

Model Program Book



SHORT TERM INTERNSHIP (On-Site/Virtual)

Designed & Developed by



**ANDHRA PRADESH
STATE COUNCIL OF HIGHER EDUCATION**
(A STATUTORY BODY OF GOVERNMENT OF ANDHRA PRADESH)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

(Onsite / Virtual)

Name of the student: KESANI NAGABABU

Name of the college: Chebrolu engineering College

Registration Number: 22HU1A0568

Period of INTERSHIP: 4 MONTHS

From: 15 -12 - 2025

To: 14 - 04 - 2025

Name and Address of the Community / Habitation: BLACKBUCKS

INTUK University YEAR

An Internship Report on

(open Enterprise Data-platform) By Data Engineering & Big Data Specialist

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

M. MOUNIKA

Department of

COMPUTER SCIENCE AND ENGINEERING

R V Institute Of Technology

Submitted by:

KESANI NAGABABU

Reg.No: 22HU1A0568

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education <https://apsche.ap.gov.in>

1. It is mandatory for all the students to complete 4 months of long-term internship either physically or virtually.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/she will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
 - a. Data and Information you are expected to collect about the organization and/or industry.
 - b. Job Skills you are expected to acquire.
 - c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

13. Never be hesitant to ask questions to make sure you fully understand what you need to do your work and to contribute to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with. 18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behaviour during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I **KESANI NAGABABU** a student of **CSE (General)** Program, Reg. No. **22HU1A0568** of the

Department of **Computer Science Engineering in chebrolu engineering college** College
do

hereby declare that I have completed the mandatory internship from 15/12/2025 to
14/04/2025

in **BLACKBUCKS** under the Faculty Guideship of **M. MOUNIKA**, Department of **Chebrolu
engineering college .**

(Signature and Date)
KESANI NAGABABU

Official Certification

This is to certify that **KESANI NAGABABU** Reg. No. **22HU1A0568** has completed his/her Internship in **BLACKBUCKS** on **Data Engineering & Big Data Specialist** under my supervision as a part of partial fulfilment of the requirement for the Degree of **B.TECH** in the Department of **COMPUTER SCIENCE AND ENGINEERING.**

This is accepted for evaluation.

(Signatory with Date and Seal)

Endorsements

Faculty Guide: M. MOUNIKA

Head of the Department: Dr.Vunnava Dinesh Babu (hod),

Principal: Dr. R.V. Krishnaiah.

Certificate from Intern Organization

**N.E.A.T**
प्रौद्योगिकी के लिए राष्ट्रीय शैक्षणिक सहयोग
National Educational Alliance for Technology

**BLACKBUCK**
EDUCATION



BLACKBUCKS EDUCATION INTERNSHIPS 2026

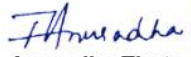
Certificate of Completion



Certificate Id: BBEDAPSCHE2026LT00371

This is to certify that **Mr/Ms Kesani Nagababu**, bearing Reg. No: **22HU1A0568**, from **Chebrolu Engineering College, Guntur**, has successfully completed a **6 Months Long-Term Internship** in **Data Engineering & Big Data Specialist** during the period **16th October 2025 to 15th April 2026**.

During the course of this internship, the candidate actively participated in the structured training sessions, practical assignments, and project-based learning activities designed to enhance industry-relevant skills and professional competencies. This internship program was conducted and **organized by Blackbucks Education Pvt. Ltd.**, with the objective of providing practical exposure and hands-on experience aligned with current industry standards.


Anuradha Thota
Chief Executive Officer
Blackbucks Education Pvt. Ltd.



Date: 01/04/2026
Place: Hyderabad



Acknowledgements

I am indebted to our Principal DR.R.V.Krishnaiah and the management of R V Institute Of Technology for providing an environment with all the facilities that helped me in completing the internship.

I am extremely grateful to Dr.Vunnava Dinesh Babu, HOD of the Computer Science and Engineering Department for his moral support and encouragement.

I wish to express my sincere gratitude to my internal guide M. MOUNIKA from the COMPUTER SCIENCE AND ENGINEERING Department, for their guidance and suggestions.

I thank all the teaching and non-teaching staff of the Department of Computer Science and Engineering and COMPUTER SCIENCE AND ENGINEERING for their kind help.

I also thank the company **Blackbucks** Team, for providing me with this golden opportunity to do the internship on the topic of Data Engineering & Big Data Specialist which helped me in learning new things.

Last but not least, I would like to add some personal notes. If there is a driving force that keeps me going, and that has not changed, it is the constant support and blessing of my parents, family, and friends. There is no doubt, in spite of my strenuous efforts; errors might remain in the internship report. Naturally, I take full responsibility for any lack of clarity, occasional erratum,

or

inexactness that may occur.

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CHAPTER1: EXECUTIVESUMMARY

The internship report shall have only a one-page executive summary. It shall include five or more Learning Objectives and Outcomes achieved, a brief description of the sector of business and intern organization and summary of all the activities done by the intern during the period.

Program Objectives :

TheBlackbucksSkillInternshipProgram2026aimsto:

1. Developindustry-readyskillsthroughstructured, domain-specifictraining.
2. Provide hands - on project experience and mentor-driven guidance. 3.
3. Strengthenplacementandemployabilityreadinessthroughmocktestsandcontinuousevaluation.
4. Offerexposuretomoderntechnologiesandworkflowsinreal-worldproblem-solving. 5.
5. Enableparticipantstograduatewithaprojectportfolioandcertification validating240hoursoflearning.

Overall Phase – Wise & Week – Wise Outcomes:

- Structuredprogressionfromdatastoragefundamentalstoadvanceddatapipelinedevelopment
- Continuous assessments and skill benchmarking
- Hands – on exposure to Hadoop, Spark, Kafka and Airflow
- Practical experience with ETL work flows and cloud data engineering
- Strongplacement-orientedpreparationandimprovedprofessionalreadiness

CHAPTER2:OVERVIEW OF THE ORGANIZATION

Suggestive contents

- A. Introduction of the Organization
- B. Vision Mission and Values of the Organization
- C. Policy of the Organization, in relation to the intern role
- D. Organizational Structure
- E. Roles and responsibilities of the employees in which the intern is placed.
- F. Performance of the Organization in terms of turnover, profits, market reach and market Val us.
- G. Future Plans of the Organization.

About the Blackbucks:

The Blackbucks : Pioneering Campus Success

The Blackbucks is India's premier Campus Success Platform, committed to transforming higher education and placement ecosystems. As a trusted partner of top Higher Education Institutions(HEIs), Blackbucks provides a robust suite of technology - driven solutions designed to empower students through rigorous training, structured practice, and successful placement outcomes.

Vision: Empowering Students for a Brighter Tomorrow

At The Blackbucks, we believe in leveraging technology to unlock the full potential of every student. Our mission is to bridge the gap between academic learning and real-world careers by offering comprehensive support systems that align education with employability.

Key Offerings:

Tap Tap Platform–Advanced learning and placement tracking system

VPL Platform - College domain wise labs

BBX Swift –PersonalizedlearningengineforupskillingandcertificationUniversity Certification

Programs – Job-linked career pathways integrated with higher education

Together,thesetoolsprovidea360°learningandplacement experience, tailored for diverse academic backgrounds.

12YearsofImpact&Innovation

Founded in 2013, The Blackbucks began with are evolutionary idea: “Technology for Students.” Over the past decade, we regrown into a world – class technology company with a track record of innovation, impact, and measurable success in student outcomes.

CHAPTER3:INTERNSHIPPART

Description of the Activities/Responsibilities in the Intern

Organization during Internship, Which shall include- details of working conditions, weekly work schedule, equipment used, and tasks performed. This part could end by reflecting on what kind of skills the intern acquired.

Module1:IntroductiontoDataEngineering

- Over view of Data Engineering & Roles
- Big Data Lifecycle
- Batch vs Streaming Architectures
- Data Engineering Use Cases

Module5:Hadoop &Data Warehousing

- HDFS Architecture & Replication
- MapReduce Workflow
- Hive Data Warehousing & HiveQL
- HBase for No SQL Storage

Module2:DataStorage&Formats

- Structured vs Unstructured Data
- Data Formats (CSV, JSON, Parquet, Avro)
- Data Schema Concepts
- Data Storage Systems Overview

Module6:ApacheSparkFundamentals

- Spark Architecture &Components
- RDDs, Transformations &Actions
- Spark Data Frames & Spark SQL
- Query Optimization(Catalyst)

Module3:ETL&DataPipelineBasics

- Extract–Transform–Load(ETL)Concepts
- Data Cleaning Techniques
- Data Pipeline Architecture
- Introduction to Work flow Automation

Module4:Big Data Foundations

- 4VsofBigData
- Distributed Systems Concepts
- Hadoop Ecosystem Overview
- Big Data Processing Models

ACTIVITY LOG FOR THE (WEEK-1 to WEEK-4)

Day & Date	Brief description of the Daily activity	Learning Outcome	Person In-Charge Signature
Day-1	Pre-Requisites Learning	Description This module focused on strengthening students' core technical and conceptual foundations required for successful participation in the internship and future industry roles. Students continued learning pending syllabus topics aligned with their selected domain.	
Day-2	Skill Foundation Module	Key Learning Outcomes Completion of balance technical topics from the assigned course Improved conceptual clarity and problem-solving skills	
Day-3	Company-Based Mock Tests	Company-based mock tests were conducted to simulate real corporate hiring assessments. These tests were designed based on actual company recruitment patterns to familiarize students with industry expectations.	
Day-4	Company-Based Mock Tests	Evaluate technical, aptitude, and logical reasoning skills	
Day-5	Company-Based Mock Tests	Students gained hands-on experience with company- style evaluations, enhancing their confidence and assessment readiness.	

WEEKLYREPORT

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Day-6	Employability Assessments	The assessments provided a structured benchmark of student readiness	
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WEEK-1(From Dt 15-12-2026 to Dt 14-02-2026

Objective of the Activity Done:

Detailed Report:

This module focused on strengthening students' core technical and conceptual foundations required for successful participation in the internship and future industry roles. Students continued learning pending syllabus topics aligned with their selected domain.

Company-based mock tests were conducted to simulate real corporate hiring assessments. These tests were designed based on actual company recruitment patterns to familiarize students with industry expectations

Evaluate technical, aptitude, and logical reasoning skills Prepare students for real placement and internship tests Improve time management and assessment handling skills Identify skill gaps and areas for improvement

Students gained hands-on experience with company-style evaluations, enhancing their confidence and assessment readiness. The assessments provided a structured benchmark of student readiness and helped track progressive improvement throughout the internship period.

The December 2025 phase of the Blackbucks Long Term Internship (LT2026) provided a structured and outcome-oriented learning experience. Through continuous learning, regular mock testing, and employability assessments, students demonstrated progressive improvement in both technical and professional competencies.

This month served as a strong foundation for advanced internship activities and real-world industry engagement in subsequent phases.

ACTIVITY LOG FOR THE WEEK(5-9)

Day & Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
Day-1	Phase Overview	The Core Concept Phase focused on modern data engineering technologies, big data frameworks, data pipeline development and cloud-based data processing. This phase strengthened practical data engineering Skills and enhanced placement readiness through continuous assessments and hands-on learning.	
Day-2	Data Storage, Formats & ETL Fundamentals	Understanding of different data storage Formats Ability to understand basic ETL workflows	
Day-3	ETL Work flows & Big Data Foundations	Ability to understand ETL pipeline architecture Basic knowledge of big data technologies	
Day-4	Hadoop Ecosystem & Distributed Data Processing	Understanding of Hadoop ecosystem component Ability to work with distributed data storage systems	
Day-5	Spark Processing & Real-Time Data Streaming	Ability to process large data sets using Spark Understanding of real-time data streaming systems	
Day-6	Work flow Orchestration & Data Lake Architecture	Improved pipeline orchestration knowledge Strengthened practical data engineering skills	

WEEK-2(From 15-02-2026 to : 14-04-2026)

Objective of the Activity Done: Detailed

Report:

The Core Concept Phase focused on modern data engineering technologies, big data frameworks, data pipeline development, and cloud-based data processing. This phase strengthened practical data engineering skills and enhanced placement readiness through continuous assessments and hands-on learning.

Learning Focus

- Structured and unstructured data concepts
- Data storage formats such as CSV, JSON ,Parquet, and Avro
- Data schema and organization concepts
- Introduction to ETL (Extract, Transform, Load)processes

ETL pipeline design and data transformation techniques

Introduction to big data ecosystems

Understanding large-scaled at a processing concepts

Data work flow and pipeline architecture

ACTIVITYLOG FORTHEWEEK(10-16)

Day & Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
Day-1	Cloud Data Engineering & Pipeline Monitoring	Ability to build ETL pipelines using Spark Understanding cloud data engineering environments	
Day-2	Big Data Processing, Streaming & Cloud Integration	Ability to design and implement scalable data pipelines Understanding of real time data processing systems	
Day-3	Data Engineering Project Development & Implementation	Ability to development complete data engineering solutions Improved integration of data pipelines and cloud services	
Day-4	Project Implementation Phase	Technical Skills ☑ Python programming ☑ SQL queries ☑ Data transformation ☑ Data pipeline design	
Day-5	Data Engineering Project Finalization & Evaluation	Ability to finalize and deliver complete data engineering project	
Day-6	Final Phase	Students successfully completed and submitted their work, show casing a well- structured portfolio aligned with Data Engineering and Big Data industry standards.	

WEEK-3(From: to:)

Objective of the Activity Done: Detailed**Report:****1. Abstract**

The Open Enterprise Data Platform is a scalable, cloud-native solution designed to handle Modern enterprise data challenges. Organizations today deal with large volumes of structured and unstructured data generated from multiple sources such as APIs, databases, and user applications. This project proposes a unified platform that enables efficient data ingestion, storage, transformation, and visualization. The system integrates tools like Airbyte for ingestion, Snowflake for storage, dbt for transformation, and BI tools for visualization. The platform supports both batch and real-time processing while ensuring scalability, reliability, and performance.

This project demonstrates a prototype implementation of the platform, including data pipelines, transformations, and analytical outputs. The final system enables better decision-making by providing clean, structured, and accessible data.

2. Introduction Background

In today's digital era, enterprises generate massive amounts of data from various systems. Managing and extracting value from this data is a major challenge.

Traditional data systems:

Are not scalable Require manual processing Lack integration

Modern enterprises need:

Automated pipelines Centralized storage Real-time analytics

Problem Statement

Organizations face difficulties in: Integrating multiple data

- ☐sources
- ☐Handling large – scale data Maintaining data
- ☐quality
- ☐Generating real-time insights

Proposed Solution

The Open Enterprise Data Platform provides: Automated data

- ☐pipelines
- ☐Cloud-based storage
- ☐Scalable transformation models
- ☐Visualization dashboards

3. Objectives

- ☐ Build a centralized data platform
- ☐ Enable real- time and batch processing
- ☐ Improve data quality and consistency
- ☐ Support business intelligence and analytics
- ☐ Ensures callability and reliability

4. Scope of the Project

In Scope

Data ingestion from CSV/API
Data storage in cloud warehouse
Data transformation using SQL Dashboard-ready output

Out of Scope

- ☐ Advanced AI model deployment
- ☐ Real-time streaming(future scope)
- ☐ Enterprise security layers

5. Literature Review Modern data platforms are based on:

- ☐ Data Ware housing concepts
- ☐ ETL/ELT pipelines
- ☐ Cloud computing

Key advancements:

- ☐ Shift from ETL→ ELT
- ☐ Use of cloud-native tools
- ☐ Automation using orchestration tools

6. System Architecture Overview

The architecture consists of four main layers:

1. Data Ingestion Layer
2. Data Storage Layer
3. Data Transformation Layer
4. Data Visualization Layer

Architecture Flow Data Sources→ Airbyte→ Snowflake → dbt → BI Tools Description of Layers

1. Data Sources
APIs Databases CSV files

2. Ingestion Layer

Extracts data from sources Loads into warehouse

3. Storage Layer

Centralized cloud storage Handles structured data

4. Transformation Layer

Clean sand processes data Applies business logic

5. Visualization Layer

Dashboards Reports

7. Tools and Technologies Air byte

- ☐ Open-source data integration tool
- ☐ Supports multiple connectors Snowflake
- ☐ Cloud data ware house
- ☐ Highly scalable dbt
- ☐ SQL- based transformation tool
- ☐ Enables modular data models Apache Airflow
- ☐ Work flow or chest ration
- ☐ Auto mates pipelines

8. System Design Data Flow Design

- ☐ Extract → Load → Transform → Analyse Module Design
- ☐ Ingestion module
- ☐ Storage module
- ☐ Transformation module
- ☐ Reporting module

Prototype Development Features

Basic ingestion pipeline SQL transformation Output generation

Testing Test Cases

Data loading test Null value handling Query validation

Performance Analysis

Fast query execution Scalable architecture Efficient data processing

Scalability Considerations

Cloud-based storage Parallel query execution Modular pipeline design

Reliability

Retry mechanisms Error logging Data validation

Security Considerations

Authentication Data encryption Access control

Challenges Faced

Data inconsistency Integration issues Schema mismatch

Future Enhancements

Real- time streaming(Kafka) Machine learning models Data governance layer

Applications

Business analytics Financial reporting Customer insights Predictive analysis

Advantages

Scalable Flexible Automated Cost-efficient

Limitations

Requires cloud setup Initial configuration complexity

CHAPTER 5: OUTCOME DESCRIPTION

Describe the work environment you have experienced *(in terms of people interactions, facilities available and maintenance, clarity of job roles, protocols, procedures, processes, discipline, time management, harmonious relationships, socialization, mutual support and team work, motivation, space and ventilation, etc.)*

The work environment I experienced during my **Data Engineering & Big Data Specialist online internship** was highly organized, collaborative, and supportive, even in a virtual setting.

People Interactions:

Communication with mentors, team leads, and fellow interns was professional and approachable. Regular virtual meetings, discussions, and feedback sessions ensured clarity and continuous learning. Team members were open to sharing ideas, which created a positive and inclusive atmosphere.

Facilities Available & Maintenance:

Since the internship was conducted online, all necessary digital tools and platforms (such as coding environments, cloud platforms, and communication tools) were well-maintained and easily accessible. Resources like recorded sessions, documentation, and datasets were consistently updated and organized.

Clarity of Job Roles:

Roles and responsibilities were clearly defined from the beginning. Tasks were assigned with proper instructions, deadlines, and expected outcomes, making it easy to stay focused and productive.

Protocols, Procedures & Processes:

The internship followed structured workflows, including task allocation, progress tracking, and submission guidelines. Standard coding practices, version control (like Git), and data handling procedures were emphasized, helping maintain consistency and quality.

Discipline & Time Management:

Strict deadlines and regular check-ins encouraged discipline. Interns were expected to manage their time effectively, balancing learning and task completion. This helped improve productivity and accountability.

Harmonious Relationships & Socialization:

Despite being remote, the environment encouraged friendly interactions through group discussions, collaborative projects, and informal chats. This helped build good rapport among team members.

Mutual Support & Teamwork:

Team work was a key aspect of the internship. Interns supported each other in solving problems, debugging code, and sharing knowledge. Mentors were always available to guide and clarify doubts.

Describe the real-time technical skills you have acquired (in terms of the job-related skills and hands on experience)

During my **Data Engineering & Big Data Specialist online internship**, I gained several real-time technical skills through hands-on practice and project-based learning.

Programming & Data Handling:

I improved my proficiency in Python for data processing, including working with libraries like Pandas and NumPy to clean, transform, and analyse large datasets. I learned how to handle structured and semi-structured data efficiently.

Big Data Technologies:

I gained practical exposure to big data tools such as Apache Spark and Hadoop. I worked on processing large-scale datasets using distributed computing concepts, which helped me understand scalability and performance optimization.

Data Engineering Concepts:

I learned how to design and implement data pipelines, including data ingestion, transformation (ETL processes), and storage. I worked with batch processing and understood the basics of real-time data streaming.

Database Management:

I acquired hands-on experience with SQL and NoSQL databases. I performed data querying, filtering, aggregation, and optimization of queries for better performance.

Cloud Platforms:

I got exposure to cloud-based services (such as AWS or similar platforms), including storing and processing data in cloud environments. I understood how cloud infrastructure supports scalable data solutions.

Version Control & Collaboration:

I used Git for version control, learning how to manage code repositories, track changes, and collaborate with team members effectively.

Data Visualization & Reporting:

I worked with basic visualization tools and libraries to present insights in a clear and meaningful way, helping in decision-making.

Problem – Solving & Debugging:

Through real-time tasks and assignments, I developed the ability to troubleshoot errors, debug code, and optimize workflows.

Describe the managerial skills you have acquired *(in terms of planning, leadership, teamwork, behaviour, workmanship, productive use of time, weekly improvement in competencies, goal setting, decision making, performance analysis, etc.*

During my **Data Engineering & Big Data Specialist online internship**, I developed several important managerial skills alongside technical knowledge.

Planning & Organization:

I learned how to plan my tasks effectively by breaking down complex assignments into smaller, manageable steps. Weekly task schedules and deadlines helped me prioritize work and stay organized.

Leadership:

While working on group tasks, I took initiative in coordinating discussions, sharing ideas, and guiding peers when needed. This helped me build confidence in leading small teams and contributing proactively.

Teamwork & Collaboration:

I worked closely with fellow interns on shared tasks, which improved my ability to collaborate, listen to others' perspectives, and contribute constructively toward common goals.

Professional Behaviour & Work Ethic:

The internship instilled discipline, punctuality, and accountability. I maintained professional communication with mentors and peers and ensured timely submission of tasks.

Workmanship (Quality of Work):

I focused on delivering high-quality outputs by carefully reviewing my work, following best practices, and continuously improving based on feedback.

Productive Use of Time:

I developed strong time management skills by balancing learning new concepts and completing assigned tasks within deadlines, avoiding delays and distractions.

Weekly Improvement in Competencies:

Through regular feedback and self-assessment, I tracked my progress each week. I consistently worked on improving both technical and soft skills.

Goal Setting:

I set short-term goals (daily/ weekly tasks) and long-term goals (mastering tools and concepts), which helped me stay focused and motivated throughout the internship.

Decision Making:

While solving tasks, I learned to make logical decisions, such as choosing appropriate tools, methods, or approaches to handle data-related problems efficiently.

Performance Analysis:

I evaluated my performance based on feedback from mentors and task outcomes. This helped me identify strengths and areas for improvement, leading to continuous growth.

Describe how you could improve your communication skills (in terms of improvement in oral communication, written communication, conversational abilities, confidence levels while communicating, anxiety management, understanding others, getting understood by others, extempore speech, ability to articulate the key points, closing the conversation, maintaining niceties and protocols, greeting, thanking and appreciating others, etc.,)

During my **Data Engineering & Big Data Specialist online internship**, I identified several areas where I could further improve my communication skills and actively worked on developing them.

Oral Communication:

I improved my speaking skills by participating in team discussions and explaining my work during meetings. To enhance further, I can practice speaking more clearly and concisely, especially while presenting technical concepts.

Written Communication:

Through emails, reports, and documentation, I learned to communicate in a structured and professional manner. I aim to improve by making my writing more precise, error-free, and easy to understand.

Conversational Abilities:

Regular interaction with mentors and peers helped me become more comfortable in conversations. I plan to improve by engaging more actively, asking relevant questions, and contributing ideas confidently.

Confidence Levels:

Initially, I hesitated to speak in group settings, but over time my confidence improved. I can further build confidence by practicing presentations and participating in more discussions.

Anxiety Management:

I learned to stay calm and composed during meetings. To improve further, I can prepare beforehand, practice mindfulness techniques, and focus on clear communication rather than fear of mistakes.

Understanding Others:

I developed active listening skills by paying attention to instructions and feedback. I can improve further by asking clarifying questions and ensuring I fully understand before responding.

Getting Understood by Others:

I worked on explaining my ideas in a simple and structured way. I can improve by using examples and summarizing key points to avoid confusion.

Extempore Speech:

I gained some experience in thinking and responding on the spot during discussions. To improve, I can practice speaking on random topics to enhance spontaneity.

Describe how you could enhance your abilities in group discussions, participation in teams, contribution as a team member, leading a team/activity.

During my **Data Engineering & Big Data Specialist online internship**, I recognized the importance of collaboration and identified ways to further enhance my abilities in group discussions, team participation, and leadership.

Group Discussions:

I can enhance my performance in group discussions by preparing in advance, staying updated with relevant topics, and contributing meaningful points. Actively listening to others and responding thoughtfully will help me engage more effectively. I also aim to speak more confidently and ensure my ideas are clearly understood.

Participation in Teams:

To improve my participation, I focus on being more proactive in team activities, volunteering for tasks, and consistently contributing ideas. Being reliable and completing assigned work on time will strengthen my role within the team.

Contribution as a Team Member:

I can contribute more effectively by supporting teammates, sharing knowledge, and assisting in problem-solving. Maintaining a positive attitude, respecting different viewpoints, and ensuring clear communication will help in building a strong team dynamic.

Leading a Team/Activity:

To enhance my leadership abilities, I aim to develop better decision-making and coordination skills. This includes setting clear goals, delegating tasks appropriately, and motivating team members. I also plan to improve by taking initiative, managing conflicts calmly, and ensuring that the team stays focused and productive.

Describe the technological developments you have observed and relevant to the subject area of training (focus on digital technologies relevant to your job role)

During my **Data Engineering & Big Data Specialist online internship**, I observed several important technological developments in digital technologies that are highly relevant to my field.

Cloud Computing Advancements:

Modern data engineering heavily relies on cloud platforms such as Amazon Web Services, Microsoft Azure, and Google Cloud Platform. These platforms provide scalable storage, distributed computing, and managed services, enabling efficient handling of large datasets without heavy on-premise infrastructure.

Big Data Processing Frameworks:

Technologies like Apache Spark and Apache Hadoop have evolved to support faster and more efficient data processing. Spark, in particular, enables in-memory computation, which significantly improves performance for real-time analytics.

Real-Time Data Streaming:

There is a growing shift from batch processing to real-time data processing using tools like Apache Kafka and Apache Flink. These technologies allow organizations to process continuous data streams for instant insights and decision-making.

Data Pipeline & ETL Automation:

Modern data engineering uses automated data pipelines with tools like Apache Airflow. These tools help in scheduling, monitoring, and managing workflows efficiently, reducing manual effort and improving reliability.

Data Warehousing & Lake house Architecture:

Advanced data storage solutions like cloud data warehouses and lakehouse architectures combine the benefits of data lakes and warehouses. Technologies such as Snowflake and Databricks support scalable analytics and unified data management.

AI & Machine Learning Integration:

Data engineering is increasingly integrated with machine learning workflows. Tools and frameworks are redesigned to support data preprocessing and model deployment, enabling predictive analytics and intelligent decision-making.

Data Security & Governance:

With increasing data usage, technologies focusing on data privacy, security, and governance have become essential. Features like encryption, access control, and compliance tools are now integral parts of data platforms.

Student Self Evaluation of the long - Term Internship

Student Name:	Registration No:
Term of Internship:	To:
From:	
Date of Evaluation:	
Organization Name &Address:	

Pleaser a tey our performance in the following areas:

Rating Scale: Letter grade of CGPA calculation to be provided

1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction ability with community	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Self-confidence	1	2	3	4	5
7	Ability to learn	1	2	3	4	5
8	Work Plan and organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
10	Creativity	1	2	3	4	5
11	Quality of work done	1	2	3	4	5
12	Time Management	1	2	3	4	5
13	Understanding the Community	1	2	3	4	5
14	Achievement of Desired Outcomes	1	2	3	4	5
15	OVERALLPERFORMANCE	1	2	3	4	5

Date: *Signature of the Student Evaluation by the Supervisor of the Intern Organization*

Student Name: Term of Internship: From: Date of Evaluation: Organization Name & Address: Name & Address of the Supervisor with Mobile Number	Registration No: To:
--	---

Please rate the student's performance in the following areas:

Please note that your evaluation shall be done independent of the Student's self-evaluation

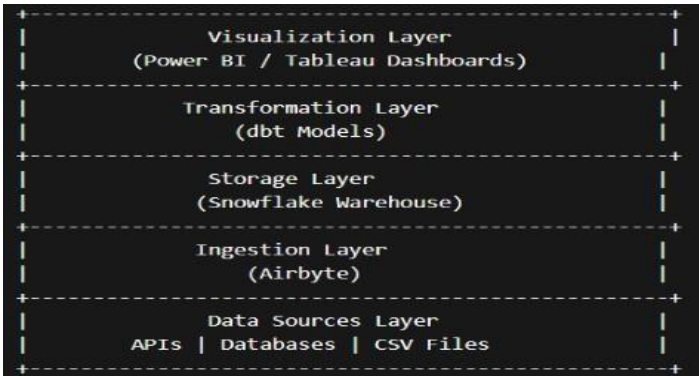
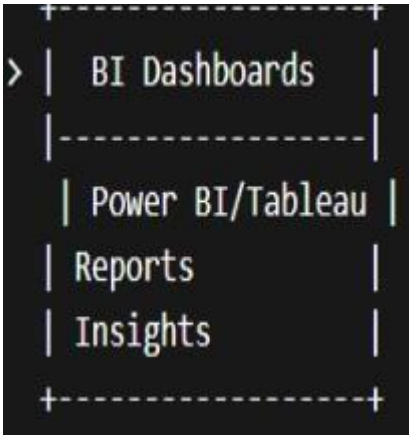
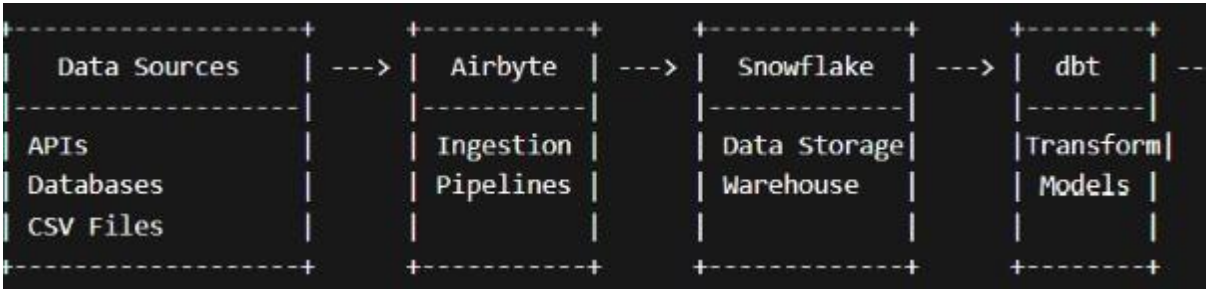
Rating Scale: 1 is lowest and 5 is highest rank

1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction ability with community	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Self-confidence	1	2	3	4	5
7	Ability to learn	1	2	3	4	5
8	Work Plan and organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
10	Creativity	1	2	3	4	5
11	Quality of work done	1	2	3	4	5
12	Time Management	1	2	3	4	5
13	Understanding the Community	1	2	3	4	5
14	Achievement of Desired Outcomes	1	2	3	4	5
15	OVERALL PERFORMANCE	1	2	3	4	5

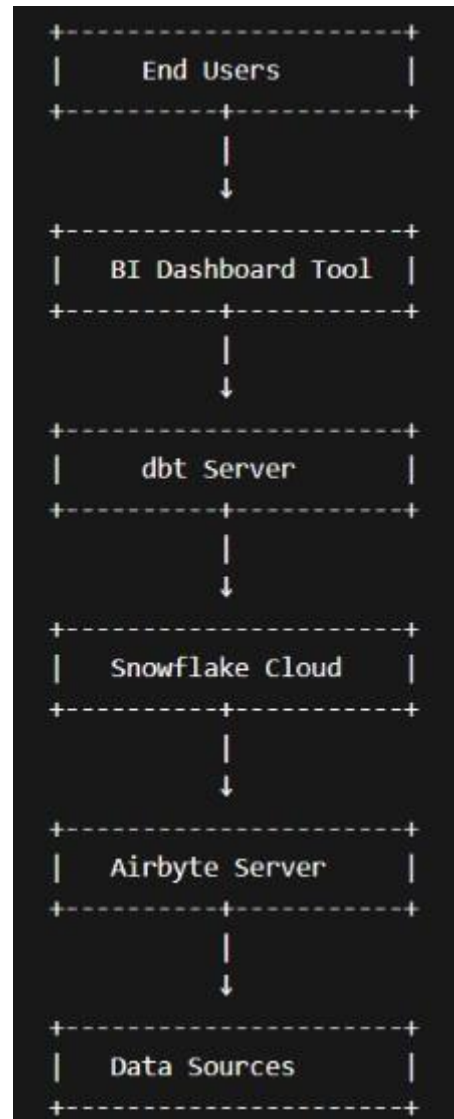
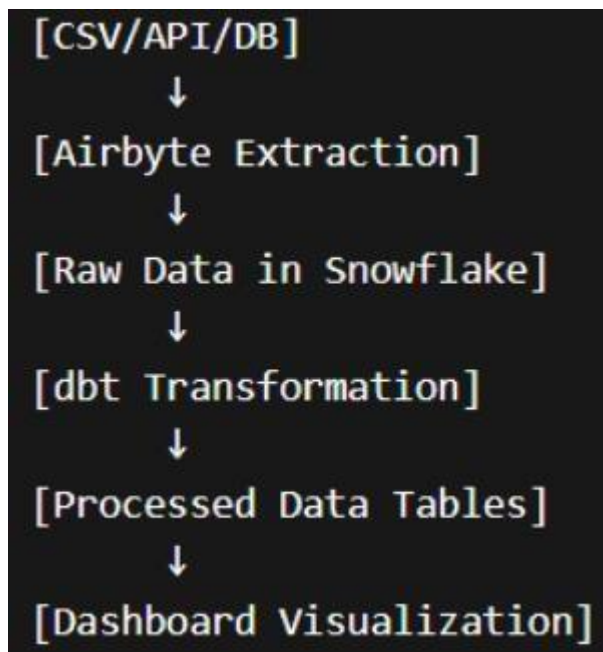
Date:

Signature of the Supervisor

PHOTOS&VIDEOLINKS



Risk	Mitigation
Data loss	Backup strategies
Pipeline failure	Retry mechanisms
Security breach	Encryption & RBAC
Data inconsistency	Validation checks



Feature	Traditional System	Modern Platform
Scalability	Limited	High
Processing	Manual	Automated
Storage	On-premise	Cloud-based
Speed	Slow	Fast
Flexibility	Low	High

Lesson Plan Link:

<https://taptap.blackbucks.me/lessonPlan/?lessonPlanId=73226&testType=collegeLessonPlan>

Python

```
import pandas as pd

data = pd.read_csv('sales_data.csv')
data.dropna(inplace=True)

print(data.head())
```

Python

```
import snowflake.connector

conn = snowflake.connector.connect(
    user='USER',
    password='PASSWORD',
    account='ACCOUNT'
)
```

SQL

```
SELECT
    customer_id,
    SUM(sales) AS total_sales
FROM sales_data
GROUP BY customer_id;
```

Sample Output

Customer	Sales
101	5000
102	3200

CSV → Python → SQL → Output

EVALUATION

Internal Evaluation for Short Term Internship(On- site/Virtual)

Objectives:

- To integrate theory and practice.
- To learn to appreciate work and its function towards the future.
- To develop work habits and attitudes necessary for job success.
- To develop communication, interpersonal and other critical skills in the future job.
- To acquire additional skills required for the world of work.

Assessment Model:

- There shall only be internal evaluation.
- The Faculty Guide assigned is in charge of the learning activities of the student and for the comprehensive and continuous assessment of the students.
- The assessment is to be conducted for 100 marks.
- The number of credits assigned is 4. Later the marks shall be converted into grade and grade points to include finally in the SGPA and CGPA.
- The weightings shall be:
 - Activity Log 25 marks
 - Internship Evaluation 50 marks
 - Oral Presentation 25 marks
- Activity Log is the record of the day-to-day activities. The Activity Log is assessed on an individual basis, thus allowing for individual members within groups to be assessed this way. The assessment will take into consideration the individual student's involvement in the assigned work.
- While evaluating the student's Activity Log, the following shall be considered—
 - a. The individual student's effort and commitment.
 - b. The originality and quality of the work produced by the individual student.
 - c. The student's integration and co-operation with the work assigned.
 - d. The completeness of the Activity Log.
- The Internship Evaluation shall include the following components and based on Weekly Report and Outcomes Description a. Description of the Work Environment.
 - b. Real Time Technical Skills acquired.
 - c. Managerial Skills acquired.
 - d. Improvement of Communication Skills.
 - e. Team Dynamics
 - f. Technological Developments recorded.

Internal Evaluation for Short Term Internship (On-site/Virtual)

Objectives:

- To integrate theory and practice.
- To learn to appreciate work and its function towards the future.
- To develop work habits and attitudes necessary for job success.
- To develop communication, interpersonal and other critical skills in the future job.
- To acquire additional skills required for the world of work.

Assessment Model:

- There shall only be internal evaluation.

- The Faculty Guide assigned is in-charge of the learning activities of the students and for the comprehensive and continuous assessment of the students.
- The assessment is to be conducted for 100 marks.
- The number of credits assigned is 4. Later the marks shall be converted into grades and grade points to include finally in the SGPA and CGPA.
- The weightings shall be:
 - Activity Log 25 marks ○ Internship Evaluation
 50marks
 - Oral Presentation 25 marks
- Activity Log is the record of the day-to-day activities. The Activity Log is assessed on an individual basis, thus allowing for individual members within groups to be assessed this way. The assessment will take into consideration the individual student's involvement in the assigned work. □ While evaluating the student's Activity Log, the following shall be considered –
 - a. The individual student's effort and commitment.
 - b. The originality and quality of the work produced by the individual student.
 - c. The student's integration and co-operation with the work assigned.
 - d. The completeness of the Activity Log.
- The Internship Evaluation shall include the following components and based on Weekly Reports and Outcomes Description
 - a. Description of the Work Environment.
 - b. Real Time Technical Skills acquired.
 - c. Managerial Skills acquired.
 - d. Improvement of Communication Skills.
 - e. Team Dynamics
 - f. Technological Developments recorded.

MARKS STATEMENT

(To be used by the Examiners)

INTERNAL ASSESSMENT STATEMENT

Name Of the Student:

Programme of

Study: Year of

Study:

Group:

Register

No/H.T. No: Name

of the College:

University:

<i>Sl.No</i>	<i>Evaluation Criterion</i>	<i>Maximum Marks</i>	<i>Marks Awarded</i>
1.	Activity Log	25	
2.	Internship Evaluation	50	
3.	Oral Presentation	25	
	GRAND TOTAL	100	

Date:

Signature of the Faculty

Guide

Certified by

Date:

**Signature of the Head of the
Department/Principal**

Seal: